



Descriptive Statistics

Part A: TYPES AND DISPLAYING DATA

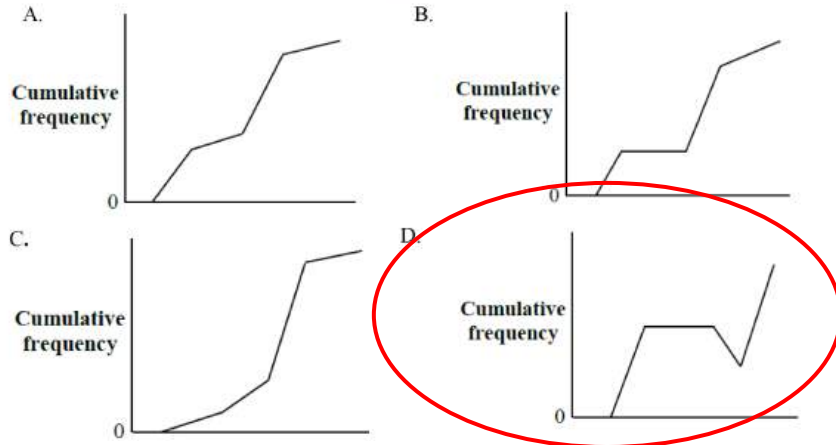
2 unit



Q 1

Hurlstone

Which of the following CANNOT be a cumulative frequency polygon?

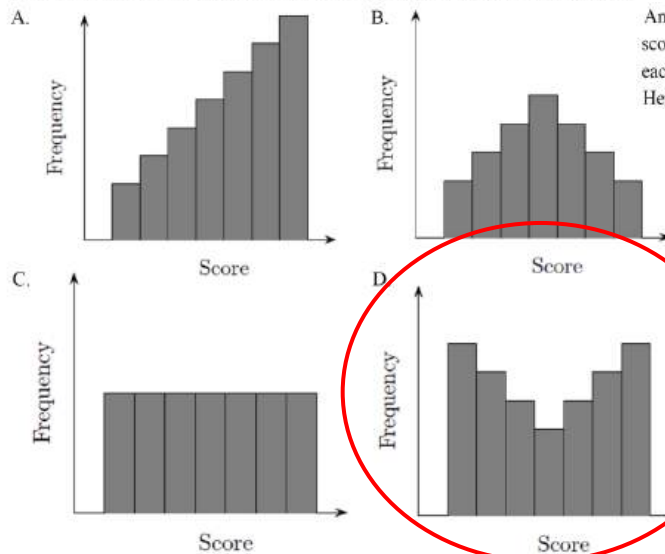


Cumulative frequency is never going to be a decreasing function.

Q 2

Hurlstone

Which of the following graphs shows data with the largest standard deviation?

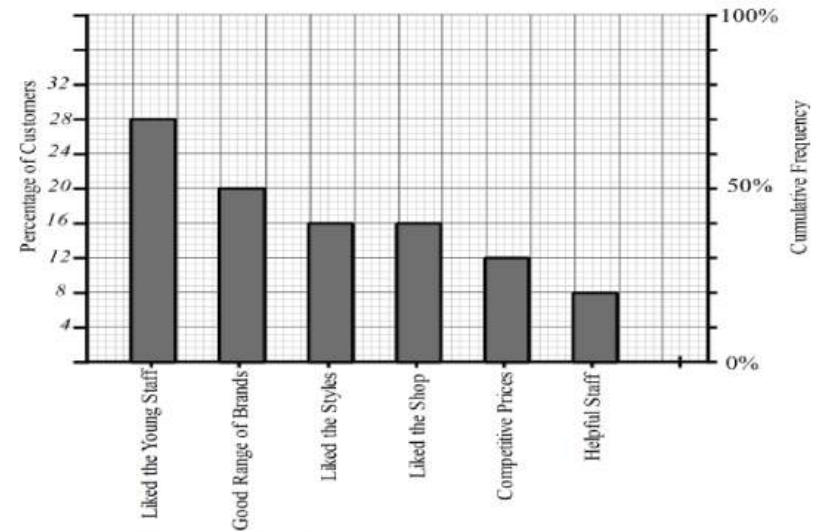


Answer: D due to a lot of low scores and high scores the distance between the mean and each of the individual scores will be greater. Hence greater Standard deviation.

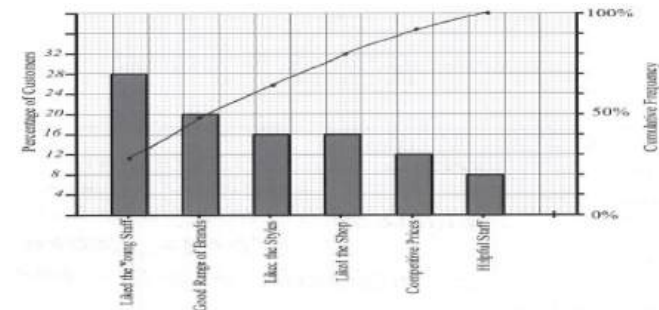
Q 3

Hurlstone

The chart below shows the reasons that 25 customers gave for shopping at a local clothing store.



Draw the Pareto line on the chart above.



Steps for Calculating Cumulative Frequency to draw the Pareto Line.

- 1) 28% $\Rightarrow$ Cumulative Frequency = 28%
- 2) 20% $\Rightarrow$ Cumulative Frequency = 48%
- 3) 16% $\Rightarrow$ Cumulative Frequency = 64%
- 4) 16% $\Rightarrow$ Cumulative Frequency = 80%
- 5) 12% $\Rightarrow$ Cumulative Frequency = 92%
- 6) 8% $\Rightarrow$ Cumulative Frequency = 100%





Descriptive Statistics

Part A: Types and Displaying Data

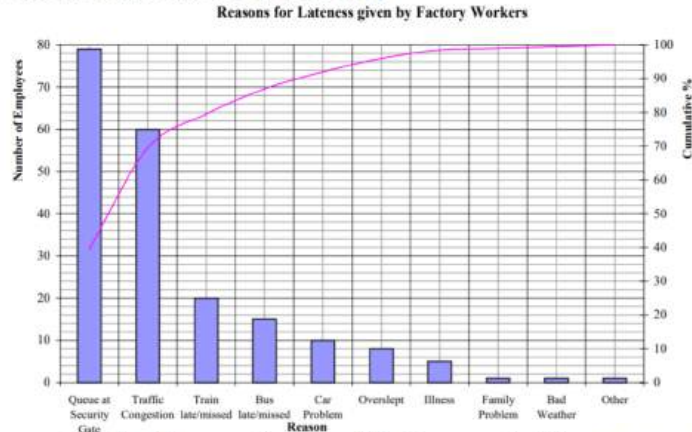
2 unit



Q 4

Girraween

A company collected data on the reasons for given by 200 factory workers for arriving late. The pareto chart shows the data collected.



What percentage of workers gave the reason "Traffic congestion"?

- A. 60% B. 75% C. 50% D. 30%

$$\text{Total} = 200 ; \text{Traffic} = 60 \\ = \frac{60 \times 100}{200} = 30\%$$

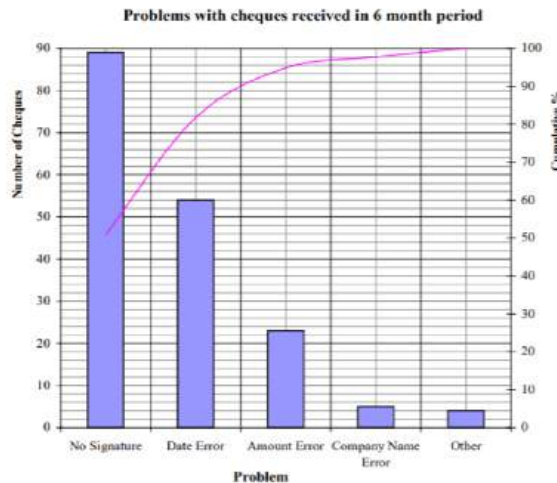
Q 5

Sydney Girls

The accounts department of a company records problems with cheques received from customers over a six-month period. The results are shown in the Pareto chart below.

Approximately what percentage of the problems are due to an Amount Error?

- A. 23%
B. 5%
C. 95%
D. 13%



Q 6

Blacktown

John works in a cake shop, and based on sales over two weeks, he conducted a survey of the five most popular cakes. What type of data is this?

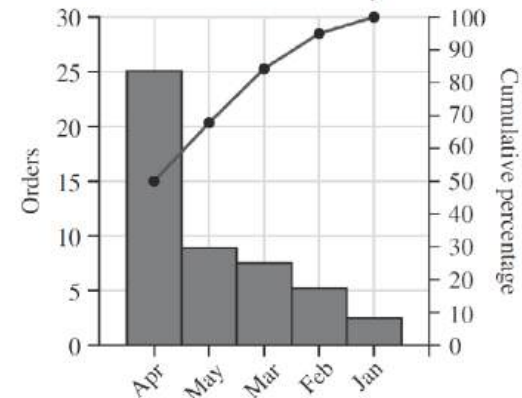
- A. Categorical nominal
B. Categorical ordinal
C. Quantitative continuous
D. Quantitative discrete

Type of cakes is categorical nominal

Q 7

Blacktown

The Pareto chart below shows the order received by a business for five months.



What percentage of orders were received in May?

- A. 69% B. 45% C. 30% D. 18%

$$68\% - 50\% = 18\% \text{ or } \frac{9}{25 + 9 + 8 + 5 + 3} = 18\%$$



Descriptive Statistics

Part A: Types and Displaying Data

2unit

selective

Q 8

Hurlstone

The Pareto chart shows complaints about the delivery of pizzas from a new pizza store.



Use the graph to answer the following questions:

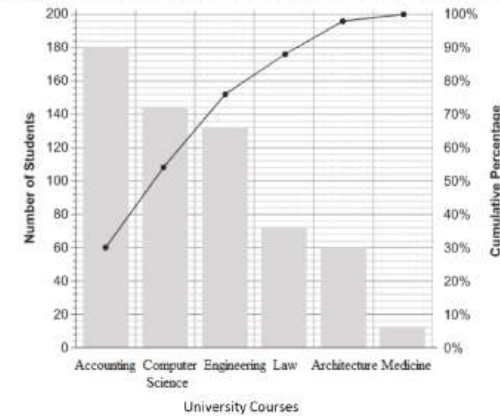
- What percentage of customers complained about late delivery of pizzas? **1**
- Explain how the Pareto principle can be used by the pizza company to direct their efforts to improve the customer experience. **2**

a	55% - 63% (acceptable range)
b	The Pareto Principal states that 80% of issues come from 20% of problems. This would direct the business to focus improving delivery times and pizza temperature as they both account for 80% of complaints made and are close to 20% of all complaint types.

Q 9

Killara

The Year 12 students from Koolara High School were asked which course they were interested in studying after graduating. The results from this survey are shown below.



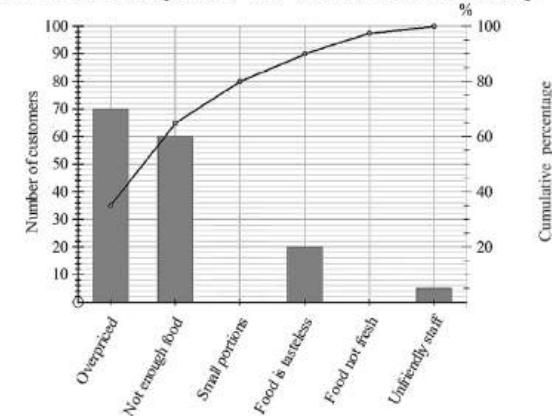
What was the total number of students who were surveyed?

- (A) 180 (B) 200 (C) 300 (D) 600

Q10

North Sydney Girls

The following Pareto chart shows 200 customer complaints from a restaurant survey. The columns for 'small portions' and 'food not fresh' are missing.



How many customers complained about the restaurant having 'small portions'?

- A. 15 people C. 40 people
B. 30 people D. 80 people



Descriptive Statistics

Part A: Types and Displaying Data

2unit



Q11

Western

As winter progresses the virus spreads much more and the health authorities decide they want to stop the virus and have been given a new medication to trial. The two-way table shows the number of people in a trial.

	Taking Medication	Control Group
Virus	204	205
No Virus	212	209

- (i) What percentage of people in the trial had the virus? 1
- (ii) What percentage of people in the control group had the virus? 1
- (iii) Determine if it is worth the health authorities using this new medication. 1

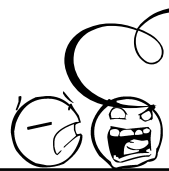
(i) Total people = $204 + 212 + 205 + 209 = 830$

Percentage of People with Virus = $\frac{204 + 205}{830} \times 100 = 49\%$

(ii) Percentage of control group with virus = $\frac{205}{414} \times 100 = 49.5\% \approx 50\%$

(iii) Percentage of medication group with virus = $\frac{204}{416} \times 100 = 49.04\% \approx 50\%$

It is not worth using the medication, since almost the same percentage of the virus occurs whether on medication or not.



Descriptive Statistics

Part B: Central Tendency and Spread

2unit

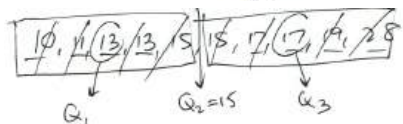
selective

Q 1

Clancy

The number of year 12 students who completed 30 minutes of exercise for each of the past ten days was as follows: 10, 11, 13, 13, 15, 15, 17, 17, 19, 28. 2

Is 28 an outlier for this set of data? Justify your answer with calculations



$$IQR = Q_3 - Q_1 = 17 - 13 = 4$$

or using calculator $IQR = 17 - 13 = 4$

$$\text{Upper limit} = Q_3 + 1.5 \times IQR = 17 + 1.5 \times 4 = 23$$

$\therefore 28$ is an outlier as it is above the upper limit of 23.

Q 2

Hurlstone

The average monthly relative humidity (in %) of city A is shown in the stem-and-leaf plot.

(i) Find the median and the inter-quartile range. 2

(ii) Draw a box-and-whisker plot to represent the data. 2

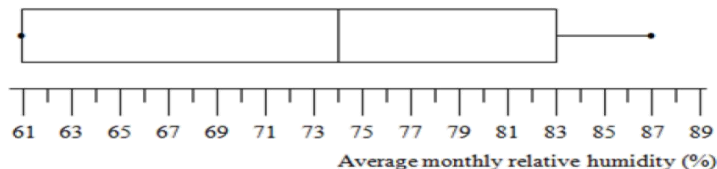
Stem	Leaf
6	1 1 1 2
7	3 5 8
8	3 7 7

(i)

$$\text{Median} = \left(\frac{73 + 75}{2} \right) \% = 74\%, \quad Q_1 = 61\%, \quad Q_3 = 83\%$$

$$\therefore \text{Inter-quartile range} = (83 - 61)\% = 22\%$$

(ii)



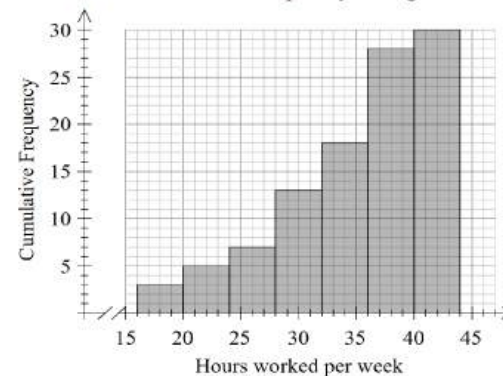
Q 3

Hurlstone

Cole is designing a survey to ask his co-workers about their job satisfaction.

One of Cole's survey questions asked how many hours each respondent works at the company.

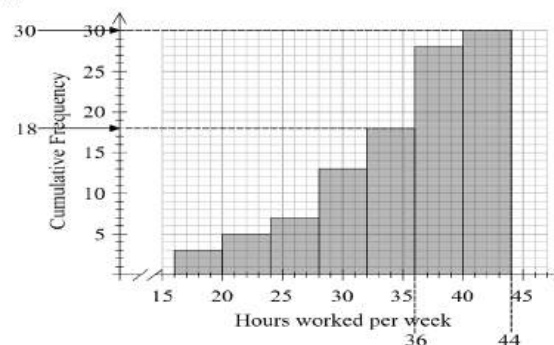
The results are shown in the cumulative frequency histogram below.



(i) What is the range of responses that gave the greatest 40% of hours worked? 1

(ii) Use the classes in the cumulative frequency histogram to estimate the mean hours worked by the respondents surveyed. 3

$$(i) 40\% \times 30 = 12$$



Top 12 respondents are from 18 to 30, which are the top two classes. These respondents worked 36 to 44 hours per week.

(ii) Class centres are the middle of each bar of the histogram. The number of respondents in each class is the change in height for each bar of the histogram.

$$\begin{aligned} \text{Sum of scores} &= 18 \times 3 + 22 \times 2 + 26 \times 2 + 30 \times 6 + 34 \times 5 + 38 \times 10 + 42 \times 2 = 964 \\ \text{Mean} &= \frac{964}{30} = 32.1 \end{aligned}$$



Descriptive Statistics

Part B: Central Tendency and Spread

Unit

selective

Q 4

Cerdon

The table below shows the total number of tries scored in 25 games of rugby.

Tries Scored	Frequency	Cumulative Frequency
0	3	3
1	5	8
2	8	16
3	3	19
4	1	20
5	1	21
6	4	25

- (a) State the range for this data. (1 mark)
 (b) Find the mean number of tries scored. (1 mark)
 (c) Find the median number of tries scored. (1 mark)

a). Range = $6 - 0 = 6$

b). Mean = 2.52

c). Median = 2

Q 5

Cambridge

The heights, in cm, of a squad of 10 basketballers are listed below in ascending order:

178, 180, 183, 183, 184, 185, 187, 187, 190, 198

- (a) What is the median height?
 (b) Find Q_1 , Q_3 and hence the interquartile range.
 (c) Draw a box plot of the given data.
 (d) Determine whether there are any outliers amongst the heights using the criterion

that an outlier is a score below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$.

(a) Median = $\frac{1}{2}(184 + 185) = 184.5$ [1]

(b) $Q_1 = 183$ and $Q_3 = 187$ so $\text{IQR} = Q_3 - Q_1 = 4$. [1]

(c) [1]



(d) $Q_1 - 1.5 \times \text{IQR} = 183 - 1.5 \times 4 = 177$

$Q_3 + 1.5 \times \text{IQR} = 187 + 1.5 \times 4 = 193$

So 198 is an outlier. [1]

Q 6

Gosford

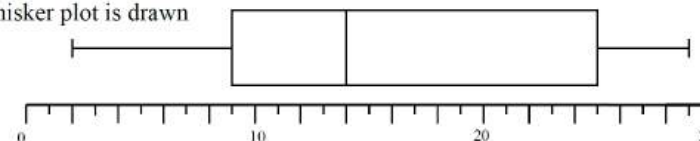
Which statement is true for an ungrouped data set with no outliers?

- (A) The largest possible range is 2 times the interquartile range.
 (B) The largest possible range is 3 times the interquartile range.
 (C) The largest possible range is 4 times the interquartile range.
 (D) The largest possible range is 5 times the interquartile range.

Q 7

Enterprise

A box and whisker plot is drawn



Which of the following is true?

- (A) Median is 14 and interquartile range is 16
 (B) Median is 9 and interquartile range is 16
 (C) Median is 14 and interquartile range is 27
 (D) Median is 14 and interquartile range is 5

Q 8

Enterprise

The times (in hours), spent listening to radio in a month for both men and women are recorded below. Determine the difference between the median scores of both the men and women.

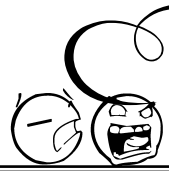
Listening to radio		
MEN	Women	
7 7 2	0	6
8 6 4 3 2	1	2 3 5 8
9 3 1 1	2	2 3 4 4 5 6 6
5 5 2	3	2 2 9 9

(A) 6 hours

(B) 8 hours

(C) 11 hours

(D) 125 hours



Descriptive Statistics

Part B: Central Tendency and Spread

2 unit

selective

Q 9

Sydney Tech

The number of students absent from Year 12 for the past nine days was as follows:

14, 17, 13, 16, 17, 12, 11, 28, 19

- Find the standard deviation. Answer correct to one decimal place.
- Find the lower quartile and the upper quartile
- Find the interquartile range

1
1
1

- (a) $\sigma_x = 4.8$
- (b) 11 12 13 14 16 17 17 19 28
 $Q_1 = 12.5$ $Q_3 = 18$
 $\therefore$ Lower quartile is 12.5
Upper quartile is 18
- (c) $IQR = 18 - 12.5 = 5.5$

Q10

Sydney Grammar

A experimenter records the following data:

4 6 8 8 9 10 12 12 17 24

- Use the standard criterion for outliers on your reference sheet to determine any outliers in the data.
- Find the mean and variance for the dataset.
- Write down a dataset with the same variance but a mean of 15.

3
2
1

- (a) $IQR = 12 - 8 = 4$
Outlier if $> 12 + 1.5 \times 4 = 18$
or if $< 8 - 1.5 \times 4 = 2$
thus 24 is an outlier
- (b) From a calculator, $\bar{x} = 11$
 $\sigma_x = 5.5$
and variance = $\sigma_x^2 = 30.4$
- (c) Add 4 to all values
8 10 12 12 13 14 16 16 21 28

Q11

Sydney Girls

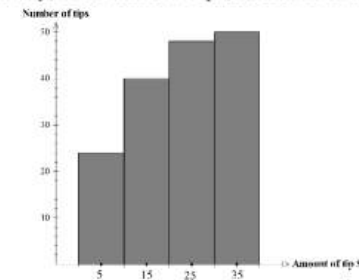
A set of data has a lower quartile of 10 and an upper quartile of 16. What is the maximum range for this set of data if there are no outliers?

- A. 24
B. 22
C. 26
D. 20

Q12

Manly

Peter works in a restaurant, and each customer can leave him a tip, some extra money, for good service. On one day, Peter earns 50 tips, as shown in the cumulative histogram:



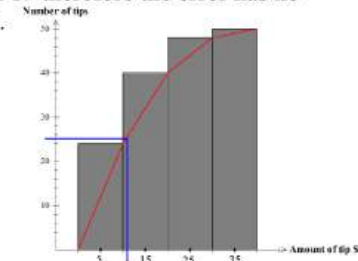
One of the tips was incorrectly listed as \$15 instead of \$25.

Giving reasons, explain what effect this would have on each of the following statistics:

- The median:
- The mean:

1
1

- a Through use of the OGIVE, it can be seen that the median is lower than 15 therefore the error has no effect on the median.



- b Since the mean calculation uses the specific value of each tip, increasing a single tip from \$15 to \$25 must increase the mean by $\frac{\$10}{50} = \0.20 .





Descriptive Statistics

Part B: Central Tendency and Spread

2 unit

selective

Q13

Grafton

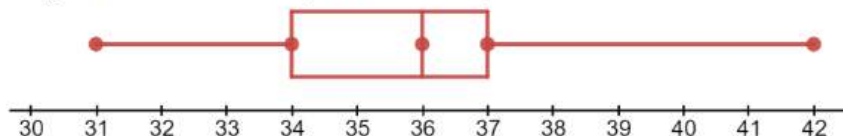
The mean of a set of data is 14 and the standard deviation is 2.1. If each score in the data set is increased by 4, which of the following statements will be true?

- (A) The mean and standard deviation will increase by 4
- (B) The mean will increase by 4 and the standard deviation will not change**
- (C) The mean will not change, and the standard deviation will increase by 4
- (D) The mean and standard deviation will increase by a factor of 4

Q14

Cabramatta

The box-plot represents the neck size, in centimetres, of 220 men.



- (a) What percentage of men has a neck size of less than 34 cm? 1
- (b) How many men have a neck size of at least 36 cm? 1
- (c) Explain if a neck size of 42 centimetres is an outlier. 1

a). 25%

b). $50\% \times 220 = 110$

c). $Q_1 = 34$ $Q_3 = 37$

$IQR = Q_3 - Q_1 = 3$

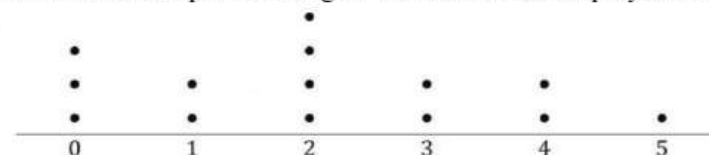
Upper bound = $Q_3 + 1.5 \times IQR = 37 + 1.5 \times 3 = 41.5$

$\therefore 42$ is an outlier.

Q15

Hunters Hill

A sample of 14 people were asked to indicate the time (in hours) they had spent watching television on the previous night. The results are displayed in the dot plot below:



Find the mean and sample standard deviation of these times. Give your answers correct to one decimal place.

total point 14.

x	0	1	2	3	4	5
f	3	2	4	2	2	1
p	$\frac{3}{14}$	$\frac{2}{14}$	$\frac{4}{14}$	$\frac{2}{14}$	$\frac{2}{14}$	$\frac{1}{14}$

$$\bar{x} = E(X) = \sum xp$$

$$= 0 \times \frac{3}{14} + 1 \times \frac{2}{14} + 2 \times \frac{4}{14} + 3 \times \frac{2}{14}$$

$$+ 4 \times \frac{2}{14} + 5 \times \frac{1}{14} \quad \textcircled{1}$$

$$= 2 \frac{3}{14} \approx 2.1$$

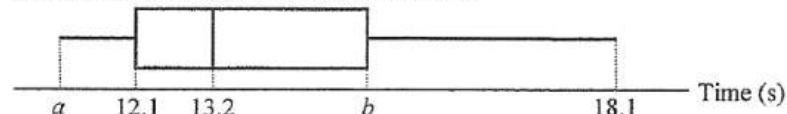
$$s = \sqrt{\text{Var}(X)} = \sqrt{E(X^2) - \mu^2} = \sqrt{0 + \frac{2}{14} + 4 \times \frac{4}{14} + 9 \times \frac{2}{14} + 16 \times \frac{2}{14} + 25 \times \frac{1}{14}} - 2.1^2$$

$$= 1.5 \quad \textcircled{2}$$

Q16

Hurlstone

The box-and-whisker diagram below shows the distribution of the times taken by a large group of students of an athletic club to finish a 100m race:



The inter-quartile range and the range of the distribution are 3.2s and 6.8s respectively. What are the values of a and b ?

- A. $a = 11, b = 15.65$
- B. $a = 11, b = 14.3$
- C. $a = 11.3, b = 15.3$**
- D. $a = 11.2, b = 15.65$

$$b - 12.1 = 3.2 \text{ and } 18.1 - a = 6.8$$

$$b = 15.3 \text{ and } a = 11.3$$



Descriptive Statistics

Part B: Central Tendency and Spread

2unit

selective

Q17

Total Education

A cumulative frequency table for a data set is shown.

- (a) What is the interquartile range of this data set?
(b) Explain if this data set contains an outlier.

(a) $\frac{1}{4}(63+1) = 16 \Rightarrow Q_1 = 3$ (16th score)

$\frac{3}{4}(63+1) = 48 \Rightarrow Q_3 = 7$ (48th score)

$IQR = Q_3 - Q_1 = 4$

(b) $Q_1 - 1.5 \times IQR = 3 - 6 = -3$

$Q_3 + 1.5 \times IQR = 7 + 6 = 13$

There is no score less than -3 or more than 13 .

Therefore data set does not contain any outlier.

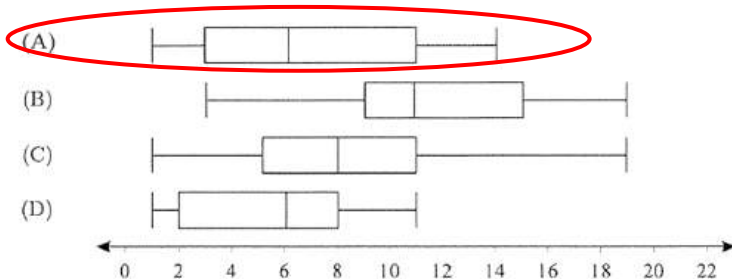
Score	Cumulative frequency
1	6
2	10
3	17
4	20
5	28
6	40
7	49
8	62
12	63

2
1

Q18

PrairieWood

Which of the data sets displayed in the following box-and-whisker plots has the largest interquartile range?



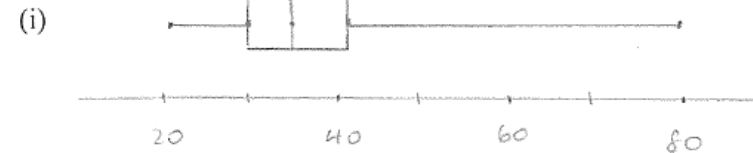
Q19

PrairieWood

The stem-and-leaf plot shows the ages of 25 actors when they received their award for the Best Actor.

Stem	Actors
2	1 4 6 6 6
3	0 0 1 1 3 3 4 4 4 5 7
4	1 1 1 1 9
5	
6	0 1
7	5
8	0

- (i) Use the data given to construct a box-and-whisker plot. 2
(ii) Does the score of 80 qualify as an outlier? Support your answer with mathematical calculations. 2



- (ii) Outlier if $80 > Q_3 + 1.5 \times IQR$
 $IQR = 41 - 30 = 11$
 $41 + (1.5 \times 11) = 57.5$
 Since $80 > 57.5$, it is an outlier.



Descriptive Statistics

Part A: Types and Displaying Data

2 unit



Q20

Ace

Class A has 24 students and achieved a mean on an assessment task of 75.5%.
Class B has 28 students and achieved a mean on the same assessment task of 80.5%. What was the mean mark for both classes. Answer correct to one decimal places.

Class A total number of marks $75.5 \times 24 = 1812$.

Class B total number of marks $80.5 \times 28 = 2254$

$$\text{Mean} = \frac{1812 + 2254}{24 + 28}$$

$$= 78.1923 \dots \%$$

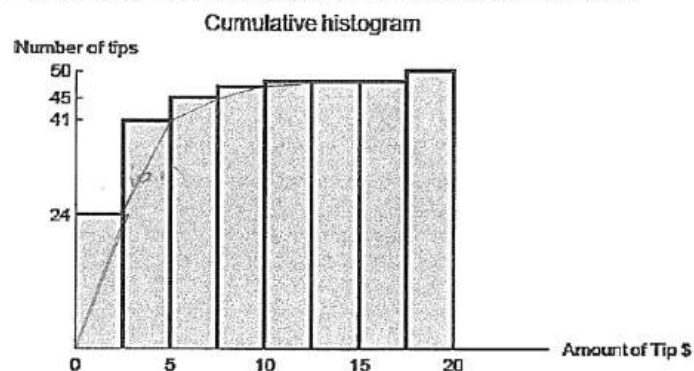
$$\approx 78.2\%$$

$\therefore$ Mean mark for both classes is 78.2%

Q21

Cecil Hills

Peter works in a restaurant, and each customer can leave him a tip, some extra money, for good service. On one day, Peter earns 50 tips, as shown in the cumulative histogram:



One of the tips was incorrectly listed as \$6, instead of \$16.

Giving reasons, explain what effect would this have on each the following statistics?

(a) The median.

2

(b) The mean.

2

(a) $= \$2.50$
Since \$6.00 is above the median already
 $\therefore$ \$16 correction would not affect the median

(b) Another \$10 would increase the value of the mean by $\frac{10}{50} = \$0.20$



Descriptive Statistics

Part C: Analysing Data

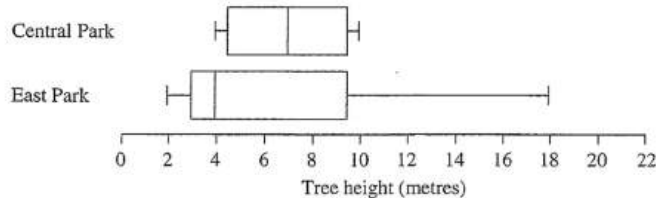
2 unit

selective

Q 1

Fort Street

Annie and her biology class went to two large city parks and measured the heights of the trees in metres. In Central Park there were 25 trees. In East Park there were 27 trees. The data sets were displayed in two box-and-whisker plots.



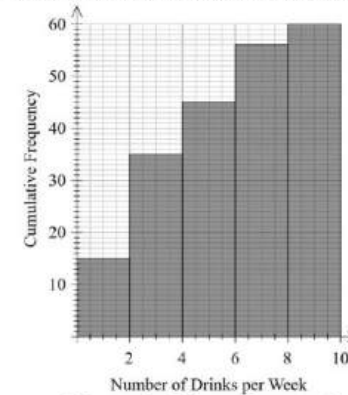
Compare and contrast the two data sets by examining the shape and skewness of the distributions, and the measures of location and spread.

1. The Central Park tree distribution is symmetrical whereas the East Park distribution is positively skewed.
2. The median height of the trees in Central Park is greater than that of East Park (7 compared to 4). In fact, we can see that the median height of the trees in East Park is the same as the shortest tree in Central Park meaning 50% of the trees in East Park are shorter than the shortest tree in Central Park.
3. East Park has a bigger range (16 compared to 6). However, their interquartile ranges are more similar (6:5 compared to 5). East Park's tree heights are slightly less consistent.

Q 2

St George Girls

Lizzie is a university student collecting data about her classmates' alcohol consumption. A question in her survey asks the respondent to indicate how many alcoholic drinks they typically consume each week. The data collected is displayed in the cumulative frequency histogram.



- How many drinks per week do each of the students in the top decile of respondents consume?
- Construct a box-plot to represent the data.



- Comment on the distribution of the data.

- The top 10% drink 7.5 to 10 drinks per week.
- Any integer between 7.5 to 10 - 1 mark
A range (7.5 - 10 or 8 - 10)
- The distribution is positively skewed - 1 mark

